
DISPROVING SOME THEOREMS IN SHARMA AND CHAUHAN *et al.* (2018, 2021)*

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ABSTRACT

The main objective of this work is to show, through counterexamples, that some of the theorems presented in the papers of Sharma *et al.* (2018) and Chauhan *et al.* (2021) are incorrect. Although they used these theorems to establish a sufficient condition for a multi-twisted (MT) code to be linear complementary dual (LCD), we show that this condition itself remains valid. We further improve this condition by removing the restrictions on the shift constants and relaxing the required coprimality condition. We show that compared to the previous condition, the modified condition is able to identify more LCD MT codes. Furthermore, without the need for a normalized set of generators, we develop a formula to determine the dimension of any ρ -generator MT code.

Keywords Multi-twisted code · linear complementary dual · Determinantal divisors · Algebraic coding

MSC: 94B05, 94B60, 11T71

1 Introduction

Multi-twisted (MT) codes over a finite field $\mathbb{F}_q$ constitute a significant and comprehensive class of linear codes. This class contains several well-known subclasses, including cyclic, constacyclic, quasi-cyclic, quasi-twisted, and generalized quasi-cyclic codes. For some integer $\ell \geq 1$, let $0 \neq \lambda_i \in \mathbb{F}_q$ and $m_i \geq 1$ for $1 \leq i \leq \ell$. If $\Lambda = (\lambda_1, \lambda_2, \dots, \lambda_\ell)$, then a Λ -MT code $\mathcal{C}$ with block lengths $(m_1, m_2, \dots, m_\ell)$ is defined in [1, Definition 3.1] as a linear code of length $n = m_1 + m_2 + \dots + m_\ell$ that remains invariant under the Λ -MT linear transformation

$$T_\Lambda : (c_{1,0}, c_{1,1}, \dots, c_{1,m_1-1}; c_{2,0}, c_{2,1}, \dots, c_{2,m_2-1}; \dots; c_{\ell,0}, c_{\ell,1}, \dots, c_{\ell,m_\ell-1}) \mapsto \\ (\lambda_1 c_{1,m_1-1}, c_{1,0}, \dots, c_{1,m_1-2}; \lambda_2 c_{2,m_2-1}, c_{2,0}, \dots, c_{2,m_2-2}; \dots; \lambda_\ell c_{\ell,m_\ell-1}, c_{\ell,0}, \dots, c_{\ell,m_\ell-2}).$$

Throughout this paper, we adopt the same notations as in [1, 2]. Thus, $\mathcal{C}$ denotes a Λ -MT code over $\mathbb{F}_q$ with block lengths $(m_1, m_2, \dots, m_\ell)$. The Euclidean dual $\mathcal{C}^\perp$ of $\mathcal{C}$ is a $(\lambda_1^{-1}, \lambda_2^{-1}, \dots, \lambda_\ell^{-1})$ -MT code with the same block lengths. By using polynomial representation for blocks, $\mathcal{C}$ can be regarded as an $\mathbb{F}_q[x]$ -submodule of the Λ -MT module

$$V = \bigoplus_{i=1}^{\ell} \frac{\mathbb{F}_q[x]}{\langle x^{m_i} - \lambda_i \rangle} = \frac{\mathbb{F}_q[x]}{\langle x^{m_1} - \lambda_1 \rangle} \oplus \frac{\mathbb{F}_q[x]}{\langle x^{m_2} - \lambda_2 \rangle} \oplus \dots \oplus \frac{\mathbb{F}_q[x]}{\langle x^{m_\ell} - \lambda_\ell \rangle}.$$

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